Harshad Thok

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Education

Pimpri Chinchwad College of Engineering, Pune Aug 2025 – Present
Bachelor of Technology in Computer Engineering CGPA - *8.55 /10*

K K Wagh Polytechnic, Nashik Sep 2022 -Jun 2025
Diploma in Computer Technology *PERCENTAGE– 94.57%*

Projects

AetherOps – Autonomous Telemetry & Self-Healing Platform Jan 2026 – Mar 2026
(React, Vite, Tailwind CSS, Node.js, Express.js, Socket.IO, Python, FastAPI, MongoDB, Redis, Docker)

- Developed a distributed AIOps platform for real-time telemetry monitoring and anomaly detection using rolling Z-score analysis on CPU, memory, and latency metrics.
- Built backend services using Node.js and Express.js for telemetry ingestion, API management, and real-time communication.
- Integrated MongoDB for persistent storage and Redis for caching to improve data retrieval performance.
- Designed a FastAPI-based microservice for anomaly detection and alert generation.
- Created an interactive React dashboard to visualize network topology, alerts, and live system health metrics.

Railway Reservation System Dec 2025 – Jan 2026
(React, Vite, Node.js, Express.js, MySQL, CSS)

- Developed a full-stack railway reservation system with train search, seat booking, ticket generation, and PNR status tracking.
- Designed RESTful APIs using Express.js and implemented a normalized MySQL database schema.
- Built SQL queries with joins to manage route-based search and seat availability efficiently.
- Created a responsive frontend using React for improved booking experience and user interaction.
- Developed an admin module to manage train schedules, bookings, and passenger records.

Network Topology Analyzer Nov 2025 – Dec 2025
(React, Node.js, Express.js, MongoDB, Graph Visualization)

- Developed a network visualization tool to represent node connectivity and topology relationships in real time.
- Built backend APIs using Node.js and Express.js to process and manage network data.
- Implemented graph-based visualizations for topology monitoring and anomaly identification.
- Designed an interactive React interface for dynamic rendering of network insights and updates.
- Improved monitoring efficiency by enabling real-time topology analysis.

Smart Energy Management in Classroom Using IoT Jan 2025 – Apr 2025
(ESP32-CAM, PIR Sensor, Relay Module, IoT, Image Processing)

- Developed an IoT-based classroom automation system to control lights and fans based on occupancy detection.
- Integrated ESP32-CAM, PIR sensors, and relay modules for motion sensing and appliance control.
- Implemented image-based zone detection to activate electrical devices only in occupied classroom areas.
- Improved energy efficiency by reducing unnecessary power consumption through intelligent automation.

Technical Skills

Programming Languages: C, C++, Python, JavaScript, Java, PHP
Frontend: HTML, CSS, React.js, Tailwind CSS
Backend: Node.js, Express.js, FastAPI
Databases: MySQL, MongoDB, SQL
Libraries/Frameworks: NumPy, Pandas, Scikit-learn, TensorFlow
Tools & Platforms: Git, GitHub, Docker, VS Code
Core Concepts: Data Structures and Algorithms, OOP, DBMS, Operating Systems, Computer Networks

Experience

Android App Development Intern
Innovations Hub Apr 2024 – Jun 2024

- Worked on real-time Android application development using Java and XML.
- Developed and maintained application modules to improve UI responsiveness and user interaction.
- Integrated Firebase services for real-time database, authentication, and cloud messaging.
- Collaborated on project development and testing during the internship.

Achivements And Publications

- **Published Paper:** "Energy Saving in Classroom Using IoT"— Presented at Third International Conference, Nashik (2025). -certificate
- **Certification:** UDEMY – Mastering Data Structures and Algorithms using C and C++. -certificate
- Secured 1st Prize in 3rd Internation Conference Held at GGSP-2K25. -certificate
- Secured 1st Prize at National Paper Presentation Competition at KKWP-2K25. -certificate